



CSE 641 Computer Vision

Weekly Report 7

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI.

Submitted to faculty: Prof. Mehul Raval

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Aim:

For developing, training, and evaluating 3D Convolutional Neural Networks (SegResNet) for automated multi-class brain tumor segmentation using BraTS dataset, as well as building a pipeline for visualizations and metrics.

Introduction:

The precise identification of glioma sub-segmentation (necrotic core, peritumoral edema, and enhanced tumor) represents an extremely difficult process that is essential for surgical planning. For this week, our main goal was to refine our baseline 3D CNN model using the SegResNet framework provided by MONAI. Instead of focusing merely on fundamental loss measures, we enriched our evaluation pipeline by incorporating per-class Dice Similarity Coefficient (DSC) and HD95. Moreover, we designed an elaborate visualization tool that enables us to compare the output predictions with their corresponding ground truth labels across multiple MRI modalities.

Work Completed:

Swin UNETR Model Inclusion: Managed to integrate MONAI's Swin UNETR network into our current pipeline. Sorted out all the required environmental dependencies (especially einops for complex tensor operations within self-attention).

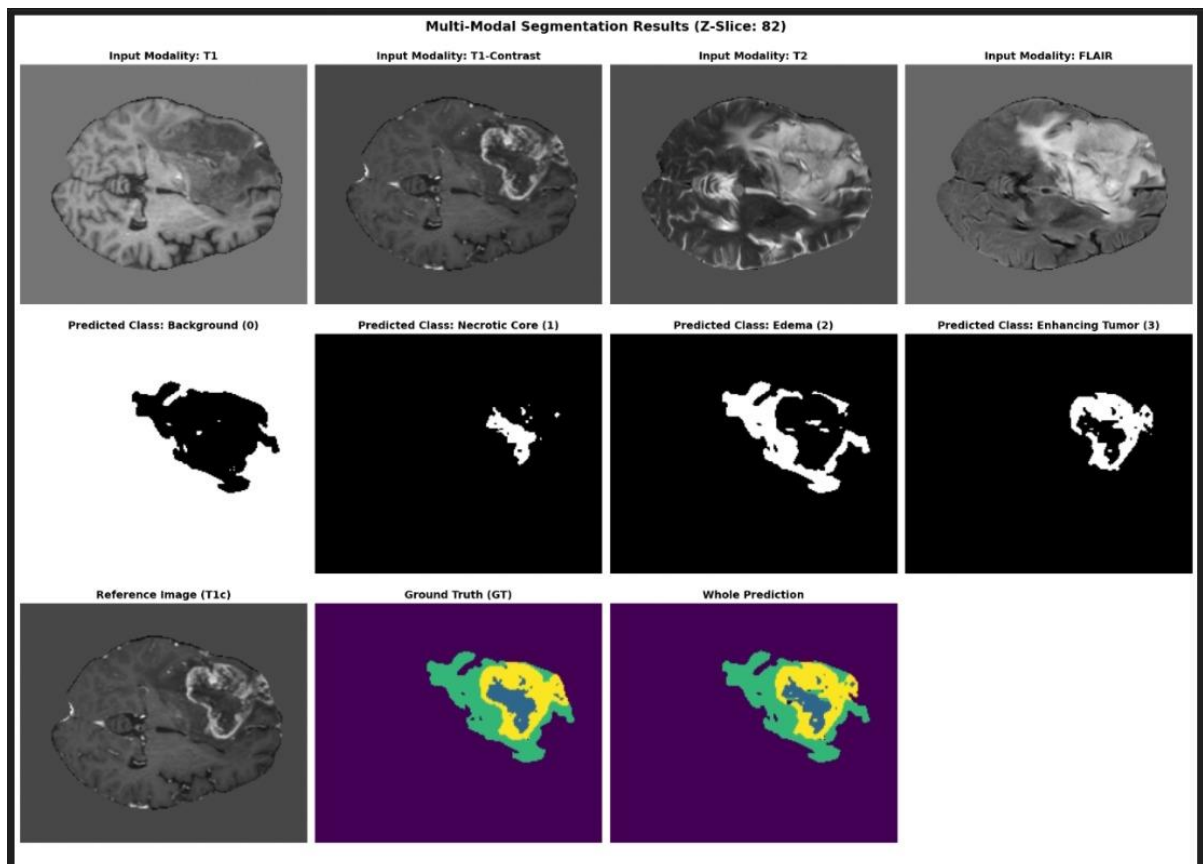
Optimizing Vision Transformer Training: Due to the increased weights of vision transformers as compared to those of CNN models, we applied gradient checkpointing and were able to train the vision transformer model on the same 8 GB GPU without any OOM issues even when we maintained our initial 96x96x96 patches.

Training Vision Transformers: From scratch, trained our Swin UNETR model using PyTorch and mixed precision training method (autocast) to lower the training epoch time as well while using the same 80:20 train/val data split to make a fair comparison with our SegResNet model from last week.

Testing & Visualizations: We ran the improved evaluation script with the newly trained weights to get the Dice Similarity Coefficient (DSC) along with the 95% Hausdorff Distance (HD95). Furthermore, created the comparisons between MRI, ground-truth, and predictions through the Swin UNETR model.

Results:

The visualization output (for example, the Z-Slice 82) shows how efficiently the model has managed to identify the general location of the tumor complex. Nevertheless, the individual segmentation masks show certain fragmentation issues specifically in the necrotic core area, indicating the limitations of the CNN structure.



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🏆 CLASS-WISE BREAKDOWN (Validation Set)
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Necrotic Core (1) - Dice: 0.5388 | HD95: 13.7866 mm
Peritumoral Edema (2) - Dice: 0.8303 | HD95: 11.7210 mm
Enhancing Tumor (3) - Dice: 0.7799 | HD95: 18.4082 mm
Overall Mean Tumor DSC: 0.7163
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Next steps and goals:

In our final report, we will compare both architectures, paying attention to how the global context in the Transformer model addresses the fragmentation of the CNN baseline predictions (particularly within the Necrotic Core). To support our findings, we will provide visualizations with our final metrics in order to prove which architecture is more appropriate for clinical use.

Conclusion:

This week ended up with an extremely solid workflow for training, evaluating, and visualizing. Our SegResNet model was benchmarked, giving a quite impressive dice score of 0.7163. What is more important is that our advanced visualization methods and metrics such as HD95 accurately identified both pros (Edema detection) and cons (fragmentation of Necrotic Core) of our model, allowing us to proceed to optimization.

References:

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5. A. Ferreira et al., "Improved Multi-Task Brain Tumour Segmentation with Synthetic Data Augmentation," arXiv preprint arXiv:2411.04632, 2024.
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